

# VIVEK MAHESH PANCHAL

Rochester, NY | 585-503-7203 | vicky.vivek14@gmail.com  
linkedin.com/in/vivek-panchal | github.com/vp1410 | vivek-panchal-portfolio

## PROFESSIONAL SUMMARY

Senior Software Engineer with 5+ years of experience building backend services, REST APIs, ETL pipelines, reporting platforms, and full-stack applications. Experienced in Python, Java, React, relational databases, AWS, Apache Airflow, Docker, and AI application architecture. Proven track record of improving system performance, reliability, data processing, and production workflows. Available immediately and open to relocation, on-site, remote, or hybrid opportunities.

## TECHNICAL SKILLS

|                                 |                                                                                                                        |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Languages:</b>               | Python, Java, JavaScript, TypeScript, PHP, SQL, Shell Scripting                                                        |
| <b>Backend and APIs:</b>        | Spring Boot, Flask, Node.js, REST APIs, Backend Services, Asynchronous Processing, Data Processing                     |
| <b>Frontend:</b>                | React, Vue.js, HTML5, CSS3                                                                                             |
| <b>Data and Workflows:</b>      | Apache Airflow, ETL Pipelines, Workflow Orchestration, Data Ingestion, Data Transformation, Reporting Systems          |
| <b>Databases:</b>               | MySQL, MongoDB, DynamoDB, Relational Database Design, Database Architecture, Indexing, Query Optimization              |
| <b>Cloud and DevOps:</b>        | AWS Lambda, API Gateway, IAM, Docker, Linux, Git, CI/CD Workflows                                                      |
| <b>Artificial Intelligence:</b> | Amazon Bedrock, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), Knowledge Bases, Conversational AI |

## PROFESSIONAL EXPERIENCE

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| <b>Senior Software Developer</b><br><i>QuinStreet, Inc.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>June 2021 – July 2026</b><br><i>Rochester, NY</i>         |
| <ul style="list-style-type: none"><li>Owned the design and development of core data and reporting infrastructure that integrated Google Ads, Meta, TikTok, Basis, LinkedIn, Bing, and other platforms into a unified analytics system.</li><li>Designed scalable Python ingestion and ETL pipelines that automated campaign-data collection and normalized cross-platform data for <b>20+ client integrations</b> with custom transformation rules.</li><li>Designed database architectures and built production forecasting, billable reporting, revenue-calculation, affiliate dashboard, and executive reporting applications using Java, Spring Boot, MySQL, and REST APIs.</li><li>Improved reporting dashboard performance to <b>2–3 second load times</b> by redesigning relational reporting tables, optimizing SQL queries, and reducing expensive aggregation operations.</li><li>Reduced data-processing workflows to <b>under 20 minutes</b> and reporting jobs to <b>under 60 minutes</b> through database indexing, query optimization, and asynchronous processing.</li><li>Designed and developed Apache Airflow DAGs to automate ETL workflows, manage dependencies, support retry handling, and improve visibility into production data-processing failures.</li><li>Developed automated monitoring and email alerts for reporting failures, unlinked campaigns, and programmatic data errors, improving production reliability and issue detection.</li><li>Built and maintained backend APIs that connected data-processing systems to React-based interfaces for forecasting, billing, revenue, and campaign-performance reporting.</li><li>Designed the end-to-end architecture for an AI live avatar agent using Amazon Bedrock, retrieval-augmented generation (RAG), Model Context Protocol (MCP), and knowledge bases to provide continuous conversational support on client websites.</li><li>Owned features from requirements analysis and architecture design through implementation, testing, deployment, documentation, and production support.</li><li>Participated in sprint planning, code reviews, and technical discussions while collaborating with engineering and business stakeholders to deliver maintainable solutions.</li></ul> |                                                              |
| <b>Full Stack Developer</b><br><i>Rochester Regional Health</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>February 2020 – December 2020</b><br><i>Rochester, NY</i> |

- Architected a serverless backend using AWS Lambda, API Gateway, IAM, and DynamoDB, improving application scalability and performance by **80%**.
- Designed database schemas and backend workflows for a healthcare innovation evaluation platform that transformed medical data into structured, actionable information.
- Established Git branching and code-management practices to improve collaboration, version-control consistency, and deployment stability.

### **Lead Web Developer**

**December 2020 – May 2021**

*Rochester Institute of Technology*

*Rochester, NY*

- Led development of a secure therapy platform using WordPress, PHP 7, and MySQL, including role-based content management, database design, and backend architecture.

### **Junior PHP Developer**

**August 2017 – February 2018**

*Network18 Media Pvt. Ltd.*

*Mumbai, India*

- Modernized legacy education platforms using a customized PHP framework, improving operational efficiency by **60%**.
- Developed backend workflows and reusable marketing-automation templates for digital publishing applications.

## **EDUCATION**

### **Master of Science in Information Science and Technology**

**2018 – 2021**

*Rochester Institute of Technology*

*GPA: 3.5/4.0*